Persistent Multi-UAV Coverage through Closed-Path Shaping with Model-Based Reinforcement Learning

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***Abstract*—Persistent coverage by multiple unmanned aerial vehicles (UAVs) requires repeated sensing of moving points of interest under partial observation while keeping online planning tractable. Direct low-level control couples sensing allocation, path geometry, and vehicle motion in a high-dimensional action space. This paper introduces a compact three-dimensional macro action that locally deforms the log-radius of a shared star-shaped closed orbit through an angular location, signed radial amplitude, and angular width. The periodic update preserves closure and positive radius, while a deterministic phase-separated follower converts the shaped path into UAV motion. Embedded in temporal difference learning for model predictive control (TD-MPC), the action enables latent-model planning over closed-path deformations. Under the same evaluation setting, the TD-MPC implementation attained mean executed coverage of 0.670, compared with 0.445 for proximal policy optimization (PPO), 0.321 for Dreamer, and 0.518 for a deterministic cyclic route based on the traveling salesman problem (TSP). Against four-UAV direct control, it achieved higher coverage (0.670 versus 0.463) at greater normalized effort per coverage. These results support closed-path shaping as a compact planning interface while revealing a coverage–effort–latency trade-off.**

***Index Terms*—persistent multi-UAV coverage, macro actions, model-based reinforcement learning, coverage path planning, partial observability**

I. Introduction

Persistent multi-UAV coverage requires a team to revisit spatially distributed targets often enough that changes are detected with limited delay. Unlike one-shot coverage, the task has no terminal covered state: points of interest (PoIs) may move, and the vehicles must continue sensing throughout the mission. Applications include wildfire observation, mobile-target surveillance, and environmental monitoring, where observations are local and the current target distribution is never fully available to the online planner. Classical persistent-surveillance methods provide strong foundations through cyclic routing, receding-horizon planning, and explicit coordination [1]–[3]. These methods also highlight a central challenge: as the target distribution changes, the planner must move the UAVs to maintain coverage, but frequent or large movements increase control effort.

Existing coverage planners represent this tradeoff at several different levels. Geometric coverage-path planning constructs vehicle paths directly from a known workspace model [4], [5]. Ergodic approaches instead drive time-averaged visitation toward a target spatial distribution and naturally support receding-horizon operation [6], [7]. Multi-target and persistent-monitoring methods add assignment, coordination, or communication constraints [8], [9]. These formulations are effective when the relevant spatial distribution can be estimated reliably, but their online optimization becomes harder when the environment is dynamic, the observations are intermittent, and the team action contains every vehicle's motion command.

Learning-based planners can use experience to infer useful behavior under uncertainty. Deep reinforcement learning has been applied to multi-robot information gathering and persistent coverage [10], [11]. Partially observable monitoring has also been addressed with belief-based policies [12], while hierarchical policies separate team-level decisions from low-level vehicle control [13]. However, the action representation remains a consequential design choice. Directly commanding the speed and turn rate of every UAV makes sensing allocation, formation geometry, and short-horizon motion part of the same decision. A dense path representation avoids some low-level coupling, but its action dimension grows with the number of path samples. Both choices enlarge the search space faced by an online model-predictive planner.

A shared closed path offers a useful intermediate representation for persistent multi-UAV coverage. It encodes persistent revisitation geometrically, provides a common reference for the team, and allows vehicles to be spaced by phase rather than independently routed at every step. Related work on coordinated circumnavigation and standoff tracking demonstrates how multiple vehicles can maintain organized motion around targets [14], [15]. For a changing PoI field, however, a fixed circle is too restrictive. Closed-path shaping must therefore adapt the coverage reference locally to observed clusters without forcing a global rescaling of the entire path. The representation should also retain closure and remain compact enough for repeated online optimization.

This paper develops a compact three-dimensional macro action for closed-path shaping. The shared coverage path is represented as a star-shaped closed orbit expressed by its log-radius over angle. At each decision step, the policy selects where to reshape the closed path, whether to expand or

contract it, and how broad the deformation should be. A wrapped Gaussian kernel converts these three variables into a smooth periodic radius update, while projection enforces path bounds. The shaping action is embedded in standard TD-MPC [16]: a learned latent model evaluates candidate action sequences and model predictive control (MPC) executes the first action. No task-specific candidate generator, residual action module, or modified planner score is required. The resulting hierarchy leaves the model-based planner responsible for closed-path shaping and leaves phase progression and bounded-speed tracking to a deterministic low-level controller.

The main contributions of this paper are:

1) A shared closed-path planning architecture addresses persistent multi-UAV coverage under partial observation.
2) A three-dimensional macro action enables smooth, bounded local reshaping of the shared closed orbit.
3) A TD-MPC-based decision-and-execution hierarchy integrates model-based path shaping with deterministic phase-separated tracking.

The rest of the paper is organized as follows. Section II formulates the partially observable persistent-coverage problem. Section III presents the closed-path shaping action and its TD-MPC integration. Section IV reports the comparative evaluation, and Section V concludes the paper.

II. Problem Formulation

A. Dynamic PoIs and Partial Observations

Let $\Omega \subset \mathbb{R}^2$ be a bounded coverage region. At time t, the environment contains N_t moving PoIs,

$$\mathrm{P}_t = \{p_{i,t} \in \Omega\}_{i=1}^{N_t}, \qquad p_{i,t+1} = f_p(p_{i,t}, \nu_{i,t}), \tag{1}$$

where $p_{i,t}$ is the position of PoI i, f_p is its motion transition map, and $\nu_{i,t}$ represents unobserved motion uncertainty. The observation o_t contains the UAV states and only the detected subset of P_t; undetected PoIs and future target motion remain hidden, requiring repeated reacquisition as illustrated in Figure 1. We summarize the finite observation history as

$$\mathrm{H}_t = o_{t-K_o+1:t}, \tag{2}$$

where K_o is the observation-history length. This history provides a finite-memory belief surrogate without assuming exact recovery of P_t. Thus the team-level decision a_t is selected only from H_t.

B. Multi-UAV Dynamics and Execution

The team contains M UAVs. For UAV j, let $x_{j,t}^u = [q_{j,t}^\mathsf{T}, \psi_{j,t}]^\mathsf{T}$ denote planar position and heading, and let $u_{j,t} = [v_{j,t}, \omega_{j,t}]^\mathsf{T}$ denote speed and turn-rate commands. For any angle α, let $\langle\alpha\rangle_\pi = ((\alpha+\pi) \bmod 2\pi) - \pi$ denote its principal value in $[-\pi, \pi)$. The executor uses bounded unicycle kinematics:

$$\begin{aligned} \psi_{j,t+1} &= \langle \psi_{j,t} + \Delta t\, \omega_{j,t} \rangle_\pi, \\ q_{j,t+1} &= \Pi_\Omega\left(q_{j,t} + \Delta t\, v_{j,t} \begin{bmatrix} \cos\psi_{j,t+1} \\ \sin\psi_{j,t+1} \end{bmatrix} \right), \end{aligned} \tag{3}$$

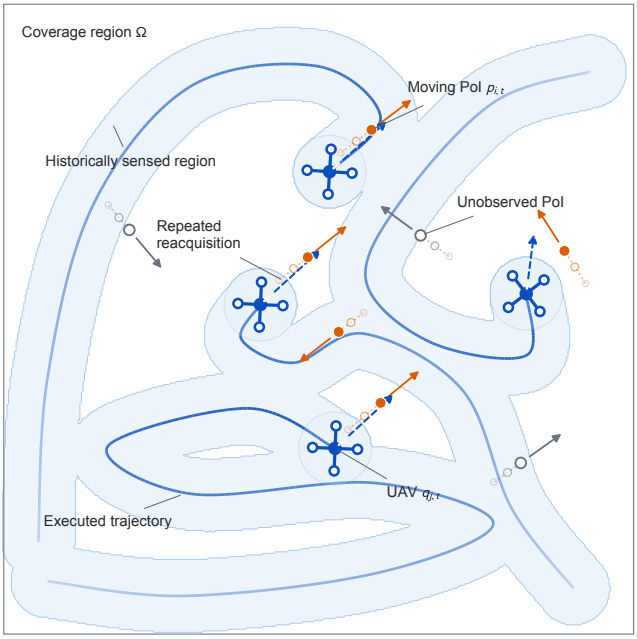


Fig. 1. Persistent multi-UAV coverage under partial observation.

where Δt is the execution interval, $0 \le v_{j,t} \le v_{\max}$, $|\omega_{j,t}| \le \omega_{\max}$, and Π_A denotes Euclidean projection onto a closed set A. The high-level decision a_t is converted into feasible commands $u_{1:M,t}$ by the deterministic path tracker described in Section III.

C. Persistent Coverage Objective

Coverage is evaluated from the persistent sensing field induced by the executed trajectories. Let $\kappa_{\sigma,r}(d) = \exp(-d^2/2\sigma^2)\mathbf{1}[d \le r]$ be a truncated Gaussian footprint, where $\mathbf{1}[\cdot]$ is the indicator function, and define $\langle f, g\rangle_\Omega = \int_\Omega f(x)g(x)\,dx$. With $\rho_{\mathrm{heat}} = e^{-\lambda_h \Delta t}$ and $S(h) = \Pi_{[0,1]}(h/\phi_{\mathrm{sat}})$, the PoI interest field, executed heat, and coverage are

$$\begin{aligned} W_t(x) &= \frac{1}{N_t}\sum_{i=1}^{N_t} \kappa_{\sigma_I, r_I}(\|x - p_{i,t}\|_2), \\ h_{t+1} &= \rho_{\mathrm{heat}} h_t + \eta_h \Delta t \sum_{j=1}^{M} \kappa_{\sigma, r_\kappa}(\|\cdot - q_{j,t+1}\|_2), \\ C_t^{\mathrm{exec}} &= \frac{\langle S(h_t), W_t\rangle_\Omega}{\langle 1, W_t\rangle_\Omega + \varepsilon}. \end{aligned} \tag{4}$$

Here σ_I and σ are the interest- and sensing-kernel widths; r_I and r_κ are their support radii; ρ_{heat} is the heat-retention factor; λ_h is the heat-decay rate; η_h is the accumulation gain; ϕ_{sat} is the saturation level; and $\varepsilon > 0$ stabilizes normalization. The hidden PoI set defines W_t only for scoring; the policy still observes only H_t. For K_p uniformly spaced reference points

$\{g_{k,t}\}_{k=1}^{K_p}$ on the nominal closed path, planned coverage is

$$\bar{\kappa}_{i,t}^{\mathrm{plan}} = \frac{1}{K_p}\sum_{k=1}^{K_p}\kappa_{\sigma,r_p}(\|p_{i,t}-g_{k,t}\|_2),$$
$$C_t^{\mathrm{plan}} = \frac{1}{N_t}\sum_{i=1}^{N_t} S\left(\frac{\eta_h M}{\lambda_h}\bar{\kappa}_{i,t}^{\mathrm{plan}}\right). \tag{5}$$

where r_p is the planned-footprint support radius. Over a horizon of T steps, the two time averages and their execution gap are

$$\bar{C}^{\mathrm{exec}} = \frac{1}{T}\sum_{t=1}^{T} C_t^{\mathrm{exec}}, \qquad \bar{C}^{\mathrm{plan}} = \frac{1}{T}\sum_{t=1}^{T} C_t^{\mathrm{plan}},$$
$$\Delta_C = \bar{C}^{\mathrm{plan}} - \bar{C}^{\mathrm{exec}}. \tag{6}$$

A positive Δ_C indicates that the nominal plan overstates realized coverage, while a negative value indicates higher soft coverage from the executed motion. Using fixed positive normalization scales v_n and ω_n, the mean quadratic control-effort surrogate is

$$U_t = \Delta t \sum_{j=1}^{M}\left[\left(\frac{v_{j,t}}{v_n}\right)^2 + \left(\frac{v_{j,t}\omega_{j,t}}{v_n\omega_n}\right)^2\right],$$
$$\bar{U} = \frac{1}{T}\sum_{t=1}^{T} U_t. \tag{7}$$

We report the effort per coverage as

$$E_{\mathrm{cov}} = \frac{\bar{U}}{\bar{C}^{\mathrm{exec}} + \varepsilon}. \tag{8}$$

This quantity supports consistent controller comparison but is not a physical battery-energy model. The task is to maximize $\mathbb{E}_\pi[T^{-1}\sum_{t=1}^{T} r_t]$ using the shaping-policy step reward

$$r_t = C_t^{\mathrm{exec}} - \lambda_u U_t, \qquad a_t \sim \pi(\cdot \mid \mathrm{H}_t), \tag{9}$$

where $\lambda_u \geq 0$ sets the coverage–effort trade-off and $\mathbb{E}_\pi$ is taken over policy-induced trajectories.

III. Methodology

A. Closed-Path Shaping with TD-MPC

We instantiate the TD-MPC planner with its second-generation implementation, TD-MPC2 [17]. Figure 2 summarizes the resulting online decision-and-execution loop. At decision step t, the partial-observation history is encoded into z_t. Policy-prior-assisted model predictive path integral (MPPI) planning then samples candidate macro-action sequences, predicts their latent evolution and returns, and selects the first action. The selected action specifies the deformation location ϕ_t, signed amplitude α_t, and angular width σ_t, which update the closed-path shape. A deterministic line-of-sight (LOS) controller projects the UAVs onto the updated path, regulates their phase gaps, and generates bounded speed and turn-rate commands. The resulting motion produces the next partial observation and closes the receding-horizon loop. TD-MPC2's world-model learning, value estimation, policy learning, and MPPI scoring retain their standard form [16], [17].

B. Closed-Path Representation

We represent the shared closed path as a star-shaped orbit with center $c_t \in \mathbb{R}^2$ and positive 2π-periodic radius $r_t(\theta)$:

$$\Gamma_t(\theta) = c_t + r_t(\theta)\begin{bmatrix}\cos\theta\\ \sin\theta\end{bmatrix}, \qquad \theta \in [0, 2\pi). \tag{10}$$

We parameterize the radius by its logarithm,

$$\ell_t(\theta) = \log r_t(\theta), \tag{11}$$

which makes positivity explicit and turns multiplicative radius changes into additive updates. The angular coordinate is periodic, so the path remains closed across the 0–2π boundary. Radius bounds $r_{\min} \leq r_t(\theta) \leq r_{\max}$ keep the reference inside the feasible path family.

C. Three-Dimensional Closed-Path Shaping Action

At each high-level step, TD-MPC2 outputs the normalized action $\tilde{a}_t = [\tilde{\phi}_t, \tilde{\alpha}_t, \tilde{\sigma}_t]^{\mathsf{T}} \in [-1, 1]^3$. Its components are mapped to the deformation location, signed amplitude, and angular width:

$$\phi_t = \pi\tilde{\phi}_t,$$
$$\alpha_t = \alpha_{\max}\tilde{\alpha}_t,$$
$$\sigma_t = \sigma_{\min} + \frac{\tilde{\sigma}_t + 1}{2}(\sigma_{\max} - \sigma_{\min}). \tag{12}$$

where $0 < \alpha_{\max} < 1$ bounds the deformation magnitude and $0 < \sigma_{\min} \leq \sigma_{\max}$ bound its angular width. The periodic deformation is defined by

$$d_{\mathbb{S}^1}(\theta, \phi) = \left|((\theta - \phi + \pi) \bmod 2\pi) - \pi\right|,$$
$$w_t(\theta) = \exp\left[-\frac{d_{\mathbb{S}^1}(\theta, \phi_t)^2}{2\sigma_t^2}\right]. \tag{13}$$

Here $d_{\mathbb{S}^1}$ is the shortest angular distance on the unit circle and w_t is the periodic Gaussian deformation weight. The closed-path shape is updated through the log radius,

$$\ell_{t+1}(\theta) = \log\left[\Pi_{[r_{\min}, r_{\max}]}\left(e^{\ell_t(\theta)}(1 + \alpha_t w_t(\theta))\right)\right]. \tag{14}$$

α_t is bounded so that $1 + \alpha_t w_t(\theta) > 0$; its sign controls local expansion or contraction, while σ_t controls the angular extent. The wrapped distance preserves continuity across the 0–2π boundary, and the projection enforces the radius bounds. The action dimension of per-UAV direct control is $2M$ for M UAVs and therefore grows linearly with team size. In comparison, the proposed action remains three-dimensional, yielding smooth, bounded, and interpretable local deformations that compose into nonuniform paths.

D. Model-Based Reinforcement Learning with TD-MPC2

TD-MPC2 encodes the partial observation stack into a latent state

$$z_t = e_\xi(o_{t-K_o+1:t}), \qquad z_{t+1} = f_\xi(z_t, a_t), \tag{15}$$

where e_ξ is the encoder, f_ξ is the latent transition model, and ξ denotes their learned parameters. From replayed H-step shaping-action sequences, it jointly learns the latent dynamics

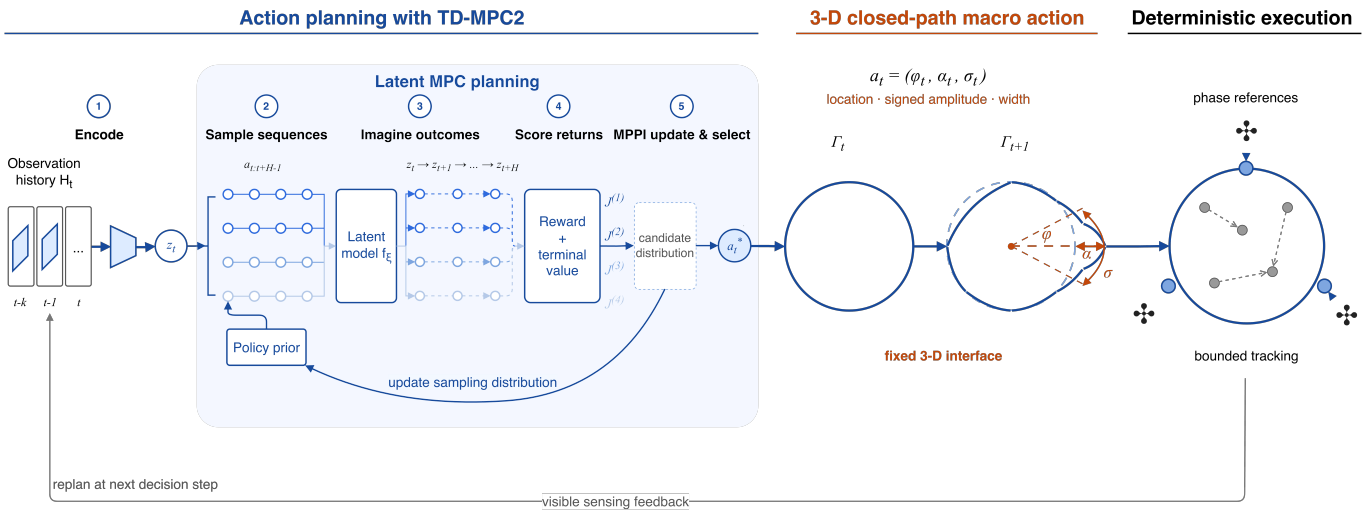


Fig. 2. TD-MPC planning with a three-dimensional closed-path macro action and deterministic execution.

f_ξ, reward model $\hat{r}_\xi$, N_Q-member action-value ensemble $\{Q_\xi^{(i)}\}_{i=1}^{N_Q}$, and policy prior π_ξ. For a candidate sequence $a_{t:t+H-1}$, the planner evaluates

$$J_{\mathrm{MPC}} = \sum_{h=0}^{H-1} \gamma^h \hat{r}_\xi(z_{t+h}, a_{t+h}) + \gamma^H \hat{Q}_\xi(z_{t+H}, \pi_\xi(z_{t+H})). \tag{16}$$

where H is the planning horizon, $\gamma \in (0, 1]$ is the discount factor, and $\hat{Q}_\xi$ is the ensemble-based terminal value estimate. Let $\bar{z}_{h+1} = \mathrm{sg}(e_\xi(o_{h-K_o+2:h+1}))$ be the stop-gradient latent target and let $\bar{\xi}$ denote the target value parameters. The bootstrapped temporal-difference (TD) target is

$$y_h = r_h + \gamma \min_i Q_{\bar{\xi}}^{(i)}(\bar{z}_{h+1}, \pi_\xi(\bar{z}_{h+1})) \,. \tag{17}$$

Here h indexes replay rollout steps, r_h is the observed reward, and $\min_i$ forms a conservative target from the value ensemble. The world model minimizes the temporally weighted objective

$$\begin{aligned} \mathrm{L}_{\mathrm{model}} = \frac{1}{H} \sum_{h=0}^{H-1} \rho_{\mathrm{TD}}^h \Big[& \lambda_c \| f_\xi(z_h, a_h) - \bar{z}_{h+1} \|_2^2 \\ & + \lambda_r \ell_r(\hat{r}_\xi(z_h, a_h), r_h) \\ & + \frac{\lambda_q}{N_Q} \sum_{i=1}^{N_Q} \ell_q(Q_\xi^{(i)}(z_h, a_h), y_h) \Big], \end{aligned} \tag{18}$$

where $\rho_{\mathrm{TD}} \in (0, 1]$ is the temporal loss-discount factor, $\lambda_c, \lambda_r, \lambda_q \geq 0$ are loss weights, and ℓ_r, ℓ_q are the reward and distributional TD regression losses. With $\bar{Q}_\xi = N_Q^{-1} \sum_i Q_\xi^{(i)}$ and entropy coefficient $\beta \geq 0$, the policy prior minimizes the temporally weighted expectation of $\beta \log \pi_\xi(a_h | z_h) - \bar{Q}_\xi(z_h, a_h)$, while the target value ensemble is updated by Polyak averaging [17].

These updates provide the three learned components used in planning: $\hat{r}_\xi$ scores each predicted step, the target-trained value ensemble supplies the terminal value, and π_ξ proposes high-value action sequences. At decision time, MPPI samples shaping-action sequences in $[-1, 1]^{3H}$, augments them with policy-prior trajectories, rolls all candidates through f_ξ, and evaluates them using (16). The highest-valued candidates update the sampling distribution; only the first shaping action is applied through (14), and planning repeats after the next partial observation. Thus, replay learning shapes both the proposal distribution and predicted return used by MPC, while the fixed three-dimensional interface keeps the online search independent of team size.

E. Deterministic Path Tracking

Let $\gamma_t : [0, L_t) \to \Omega$ be the arc-length parametrization of the updated closed path, where L_t is its total length. Each UAV is projected onto the path at $s_{j,t}$; $\Delta s_{j,t}^+$ and $\Delta s_{j,t}^-$ denote its forward and backward cyclic gaps after sorting the projected phases. A line-of-sight point is placed a distance d_{LA} ahead, giving

$$\begin{aligned} d_{j,t}^{\mathrm{LOS}} &= \gamma_t((s_{j,t} + d_{\mathrm{LA}}) \bmod L_t) - q_{j,t}, \\ \psi_{j,t}^{\mathrm{LOS}} &= \arg\big([d_{j,t}^{\mathrm{LOS}}]_1 + \mathrm{i}[d_{j,t}^{\mathrm{LOS}}]_2\big) \,, \\ v_{j,t} &= \Pi_{[0, v_{\max}]}\big(v_0 + k_c(\Delta s_{j,t}^+ - \Delta s_{j,t}^-)\big) \,, \\ \omega_{j,t} &= \Pi_{[-\omega_{\max}, \omega_{\max}]}\big(k_\psi \langle \psi_{j,t}^{\mathrm{LOS}} - \psi_{j,t} \rangle_\pi\big) \,. \end{aligned} \tag{19}$$

where $d_{j,t}^{\mathrm{LOS}}$ and $\psi_{j,t}^{\mathrm{LOS}}$ are the look-ahead displacement and heading, $[\cdot]_1$ and $[\cdot]_2$ denote Cartesian components, i is the imaginary unit, v_0 is the nominal speed, and $k_c, k_\psi > 0$ are the gap and heading gains. The gap feedback maintains phase separation, while the LOS feedback tracks the shaped path under the motion limits in (3).

The deterministic execution layer gives each learned path action stable semantics, while path projection, phase regulation, and motion limits remain outside the policy. Together with recurrent replanning, it closes the observation–shaping–tracking loop.

IV. Simulation Evaluation

A. Experiment Setup

We evaluate persistent multi-UAV coverage over four dynamic PoI scenarios, each lasting 512 high-level steps. The main comparison uses four UAVs and one frozen checkpoint for each learned method. TD-MPC2, Dreamer, and PPO all output the same three-dimensional closed-path shaping action, while the deterministic TSP baseline constructs a cyclic route through the currently observed PoIs. The executor uses a 256-point polyline with $v_0 = 0.9$, $v_{\max} = 1.8$, $\omega_{\max} = 1.5$, $k_\psi = 2.5$, $k_c = 0.15$, and $d_{\mathrm{LA}} = 0.6$; effort uses $v_n = 1.8$ and $\omega_n = 2.4$. Coverage is evaluated on a 64×64 grid with $\sigma = \sigma_I = 1.5$, $\lambda_h = 0.25$, and $\phi_{\mathrm{sat}} = 1$.

The primary metric is mean executed coverage $\bar{C}^{\mathrm{exec}}$ from (6). We additionally report normalized effort per coverage E_{cov}, mean UAV–PoI distance, signed execution gap Δ_C, and wall-clock decision time per macro step. Aggregate results are reported as mean $\pm$ sample standard deviation (SD).

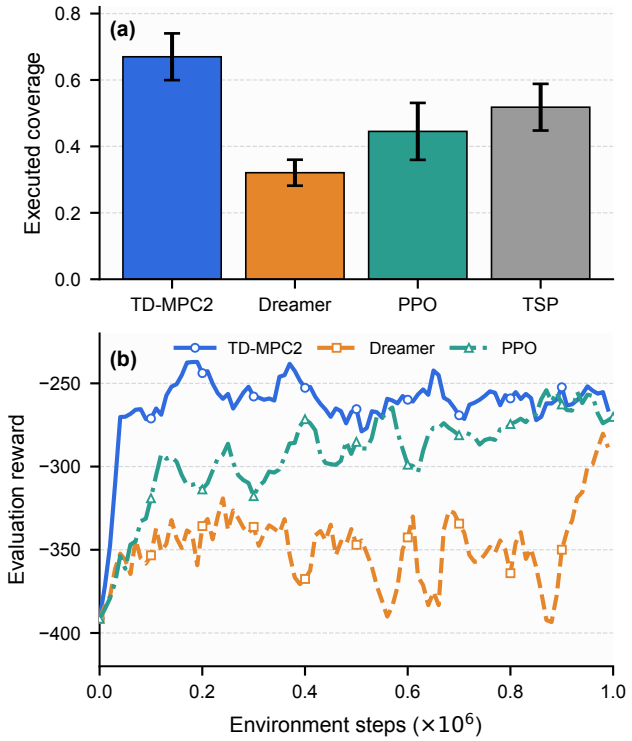


Fig. 3. Main comparison: (a) mean executed coverage with one standard deviation; (b) smoothed partially observable Markov decision process (POMDP) evaluation reward over the first 10^6 environment steps for the three learned methods.

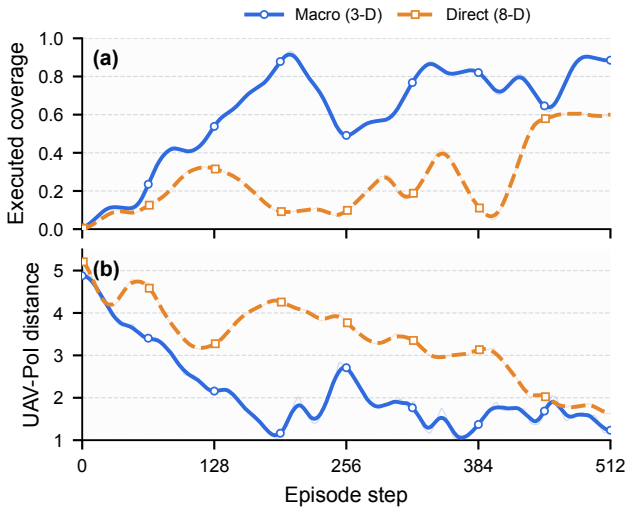


Fig. 4. Macro and Direct execution dynamics in one shared scenario: (a) executed coverage; (b) mean distance from each PoI to its nearest UAV. Thin traces are per-step values; thick traces are 17-step moving averages.

B. Comparison with Learning and Geometric Baselines

Table I shows that TD-MPC2-Macro achieved the highest mean executed coverage, 0.670 ± 0.070. Relative to TSP, TD-MPC2 increased mean coverage by 0.152 (29.3%). It also obtained the smallest mean UAV–PoI distance (2.040) and effort per coverage (0.215).

TD-MPC2 required 82.66 ms per decision, compared with 7.84 ms for Dreamer, 1.74 ms for PPO, and 0.97 ms for TSP. Thus its coverage and proximity gains came with the highest decision latency.

C. System-Level Comparison with Direct Control

We compare the closed-path shaping system with a TD-MPC2 direct-control system. Both systems use four UAVs: the direct policy outputs speed and turn rate for each UAV, giving an eight-dimensional action, whereas the shaping system uses the shared three-dimensional path action. The observation, reward, and control interfaces differ, so the results characterize the two complete systems.

Table II shows that Macro achieved higher mean coverage (0.670 versus 0.463) and a smaller mean UAV–PoI distance (2.040 versus 2.598). Direct used less normalized effort per coverage (0.161 versus 0.215) and less decision time (45.64 ms versus 82.66 ms). The comparison therefore shows a coverage–effort–latency trade-off between the two systems.

D. Representative Episode Dynamics

Over the 512-step episode in Figure 4, Macro established higher coverage earlier and kept the PoIs closer to the UAV team through most of the rollout, while both systems improved toward the end.

TABLE I
MAIN COMPARISON (MEAN ± SD)

Method	$C^{\text{exec}} \uparrow$	$E_{\text{cov}} \downarrow$	UAV–PoI Dist.↓
TD-MPC2	$\mathbf{0.670 \pm 0.070}$	$\mathbf{0.215 \pm 0.026}$	$\mathbf{2.040 \pm 0.161}$
Dreamer	0.321 ± 0.039	0.523 ± 0.068	3.639 ± 0.058
PPO	0.445 ± 0.086	0.320 ± 0.043	3.089 ± 0.537
TSP	0.518 ± 0.070	0.365 ± 0.043	2.692 ± 0.449

TABLE II
TD-MPC2 SYSTEM COMPARISON (MEAN ± SD)

System	$C^{\text{exec}} \uparrow$	$E_{\text{cov}} \downarrow$	UAV–PoI Dist.↓
Macro (3-D)	$\mathbf{0.670 \pm 0.070}$	0.215 ± 0.026	$\mathbf{2.040 \pm 0.161}$
Direct (8-D)	0.463 ± 0.059	$\mathbf{0.161 \pm 0.053}$	2.598 ± 0.269

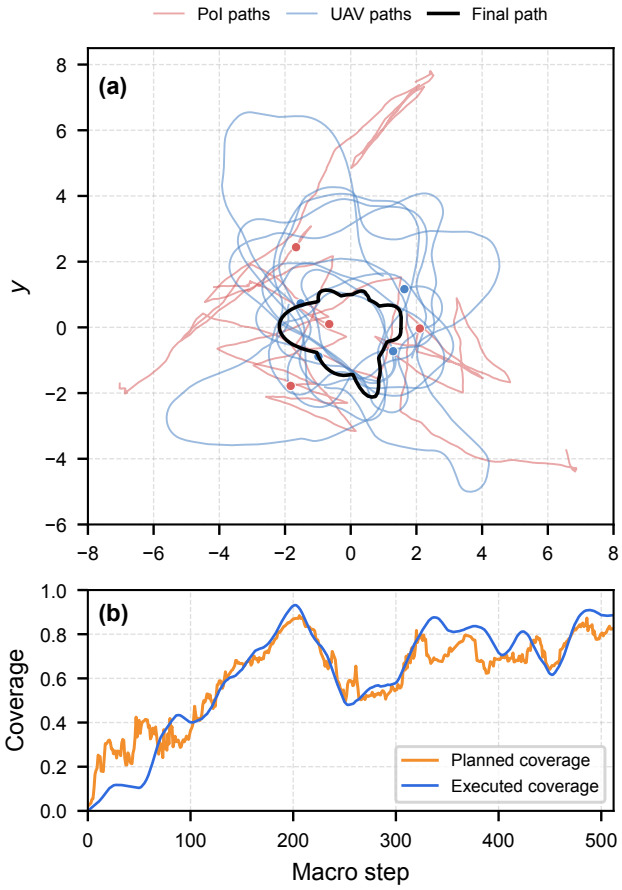


Fig. 5. TD-MPC2-Macro rollout: (a) moving PoI and UAV paths with the final shaped path; (b) planned and executed coverage over 512 steps.

In Figure 5, mean executed coverage was 0.616, mean planned-path coverage was 0.609, and the signed gap was -0.007. The small gap shows that phase-separated UAV tracking transferred the reshaped closed path into comparable sensing coverage despite moving PoIs.

V. CONCLUSION

This paper introduced a three-dimensional closed-path shaping action for persistent multi-UAV coverage. Within model-based reinforcement learning, the action gives a standard TD-MPC2 agent a compact way to reshape a shared closed path while deterministic low-level tracking converts that path into vehicle motion. TD-MPC2-Macro achieved mean executed coverage of 0.670, exceeding PPO-Macro, Dreamer-Macro, and a deterministic TSP cyclic route. The system-level direct-control comparison showed higher coverage and smaller UAV–PoI distance for the shaping system, together with greater normalized effort per coverage. Overall, closed-path shaping achieved the highest mean executed coverage among the compared methods with a fixed three-dimensional planning interface, while TD-MPC2 incurred higher decision latency.

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